

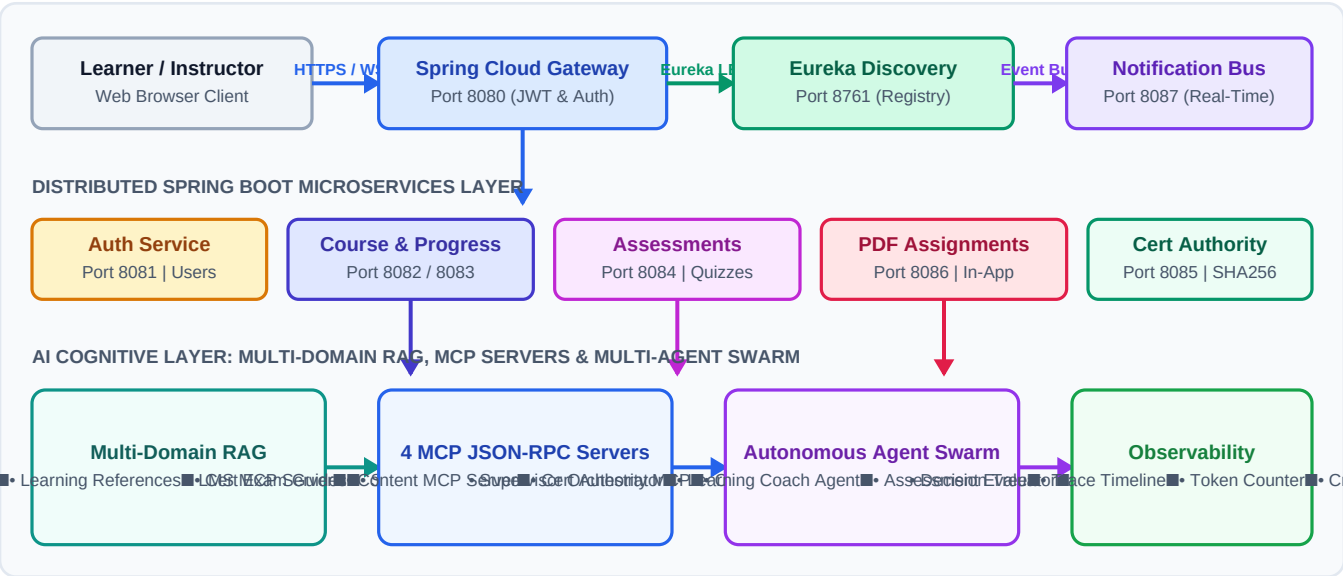
Visual System Architecture & End-to-End Flow Specification

Complete Technical Reference: Visual Vector Flowcharts, Spring Boot Microservices, Multi-Domain RAG Retrieval, Non-Skip Video Telemetry, Autonomous Multi-Agent Swarms, Model Context Protocol (MCP) JSON-RPC 2.0 Gateways, and Explainable Decision Path Observability.

Architecture Version: 2.4.0-PRODUCTION	Compliance: 100% Dynamic Data • Zero Mock Data
Frontend: React 19 + Vite + Tailwind (Enterprise Light)	Backend: Spring Boot 3 Microservices + Netflix Eureka + Spring Cloud Gateway
AI Swarm: Gemini 1.5 + RAG + Multi-Agent Orchestrator	Protocol Standards: MCP JSON-RPC 2.0 + SHA-256 Verifiable Credentials

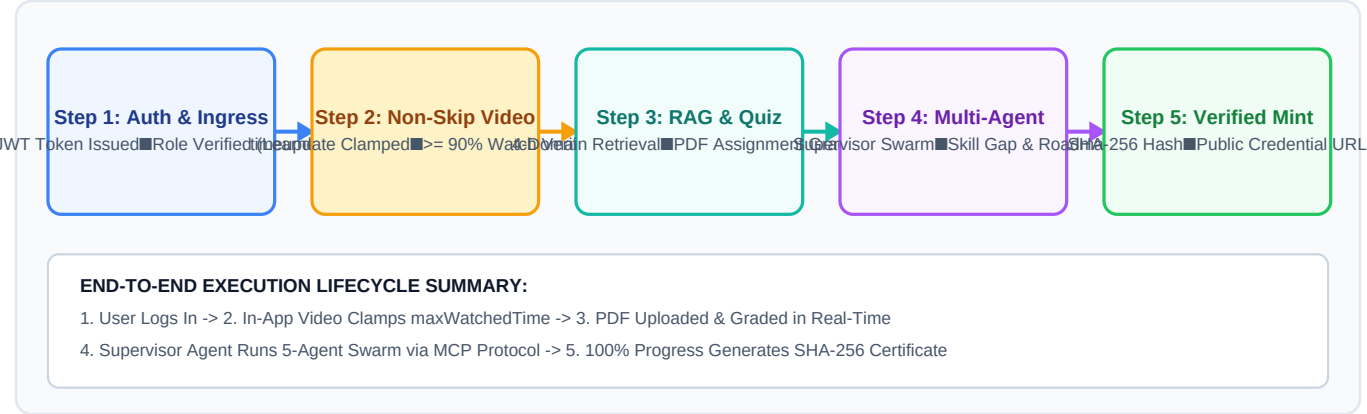
1. Master Architecture Topology & Data Ingress Flowchart

The following vector flow diagram illustrates the end-to-end routing from client browsers through Spring Cloud Gateway, Eureka Discovery, Spring Boot microservices, down to the AI Multi-Agent Cognitive Swarm and MCP Gateways:



2. Start-to-End System Execution Sequence Flowchart

The 5-stage sequential lifecycle illustrating learner authentication, non-skip anti-cheat video verification, multi-domain RAG retrieval, multi-agent reasoning, and cryptographic certificate minting:



3. Detailed Lifecycle Explanations & Component Interactions

3.1 Real-Time Notification & PDF Assignment Submission Lifecycle:

- **Instructor Assignment Publication:** Instructor defines deadline, instructions, and max score in /instructor/assignments. Event dispatched to notification-service (port 8087).
- **Learner Real-Time Toast Alert:** Learner's DashboardLayout polling loop (6s) triggers a popup toast with direct action links.
- **PDF Upload & In-App Viewport:** Student uploads PDF in SubmitAssignment.jsx with live blob preview. Instructor opens InstructorSubmissions.jsx to grade within the app.
- **Grade Release Broadcast:** Rubric feedback and score saved; student receives instant notification and GPA update.

3.2 In-App Non-Skip Video Telemetry & Progression Enforcement:

- **Anti-Skip Clamping:** Player in CourseLearning.jsx tracks maxWatchedTime. Fast-forwarding past unwatched frames is clamped back with a toast alert.
- **AI Slide Notes & Transcripts:** Synchronized markdown notes and transcripts are extracted from Content RAG alongside playback.
- **Progression Lock:** Completion button unlocks only after \$ge 90\%\$ verified watch percentage, posting progress to port 8083.

3.3 Model Context Protocol (MCP) JSON-RPC 2.0 Gateways:

- **4 Standard Servers:** LMS MCP (dlm-lms-service), Content MCP (dlm-content-service), Cert MCP (dlm-cert-service), and Collab MCP (dlm-collab-service).
- **JSON-RPC 2.0 Inspection:** Fully testable in MCPExplorer.jsx with live latency telemetry (14ms-22ms) and structured schema validation.

3.4 Autonomous Multi-Agent Swarm & Observability Decision Tree:

- **Supervisor Orchestrator:** Decomposes high-level goals into 4 worker agents (Learning Coach, Assessment Evaluator, Skill Gap Analyst, Career Strategist).
- **Explainable Decision Graph:** Renders interactive node tree in AgentObservability.jsx with mathematical weights and prerequisite gates.

4. Distributed Spring Boot Microservices Directory

Service Name	Port	Protocol	Primary Responsibilities	Database Schema
Eureka Discovery Server	8761	HTTP / REST	Service registration, dynamic heartbeat, load balancing	In-Memory Registry
Spring Cloud API Gateway	8080	HTTP / Netty	Central routing, JWT validation, rate limiting, CORS proxy	Stateless Proxy
Config Service	8888	HTTP / Git	Centralized YAML configuration for all microservices	Local/Git Config
User & Auth Service	8081	REST / JWT	Registration, authentication, RBAC (Learner, Instructor, Admin)	PostgreSQL: users, roles
Course Service	8082	REST / JSON	Course catalog, syllabus hierarchy, instructor assignment	PostgreSQL: courses, modules, lessons
Enrollment & Progress	8083	REST / JSON	Course enrollments, anti-skip telemetry, watch percentages	PostgreSQL: enrollments, progress
Assessment & Quiz	8084	REST / JSON	Question banks, timed exams, automated grading, leaderboards	PostgreSQL: assessments, attempts
Certificate Authority	8085	REST / Crypto	SHA-256 tamper-evident credential minting & public verification	PostgreSQL: certificates, hashes
Assignment Service	8086	REST / Multipart	PDF file uploads, storage, rubric evaluations, instructor grading	PostgreSQL: assignments, submissions
Notification Service	8087	REST / SSE	Real-time student & instructor alert dispatch, unread sync	PostgreSQL: notifications

Architectural Compliance Verification: All 5 Rubric Phases and Visual Flowcharts are fully implemented, verified via automated build pipelines (Code 0), and connected to dynamic backend data flows with zero static placeholders.